

Combinational ATPG

- Introduction
- Deterministic Test Pattern Generation
 - ◆ Boolean difference *
 - ◆ Path sensitization **
 - ◆ D-Algorithm [Roth 1966] **
 - ◆ PODEM [Goel 1981]**
 - ◆ FAN [Fujiwara 1983]**
 - ◆ SAT-based [Larrabee 1992]*
- Acceleration Techniques
- Concluding Remarks

*Boolean-based methods

**path-based methods

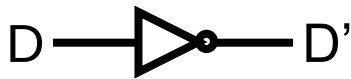
Our D-algorithm follows the original paper.
It is slightly different from WWW textbook.

D-Algorithm

- The D-algebra
- **An D-algorithm example**
- **Types of cubes**
- **Implication and Justification**
- **Flowchart of the D-algorithm**
- **Another example**
- **Problems with the D-Algorithm**

D-Algebra

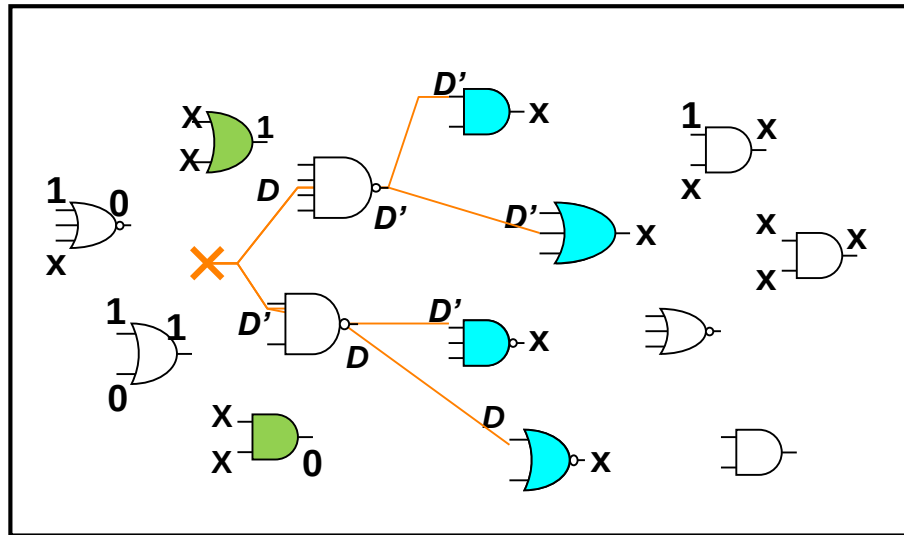
- **Five-valued logic: 1, 0, D, D', X**
 - ♦ Symbol D
 - ♦ $D = 1/0$
 - * 1 in fault-free circuit and 0 in the faulty circuit
 - ♦ $D' = \bar{D} = 0/1$
 - * 0 in fault-free circuit and 1 in the faulty circuit
 - ♦ x means “not yet specified” in ATPG



AND	0	1	D	$\bar{D}$	X
0	0	0	0	0	0
1	0	1	D	$\bar{D}$	X
D	0	D	D	0	X
$\bar{D}$	0	$\bar{D}$	0	$\bar{D}$	X
X	0	X	X	X	X

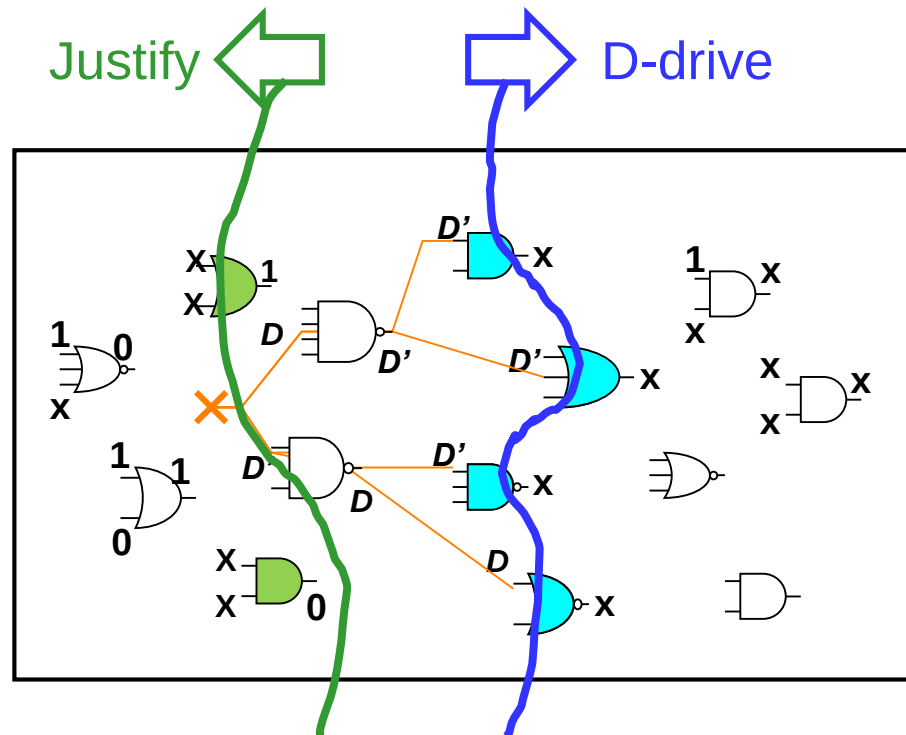
D-frontier , J-frontier

- **D-frontier** : a set of gates whose output value is currently x,
 - ♦ but have one or more D (or D') at their inputs
- **J-frontier**: a set of gates whose output value is assigned
 - ♦ But input values have not been decided yet
- Example
 - ♦ Blue gates are D-frontier; Green gates are J-frontier



Idea of D-Algorithm

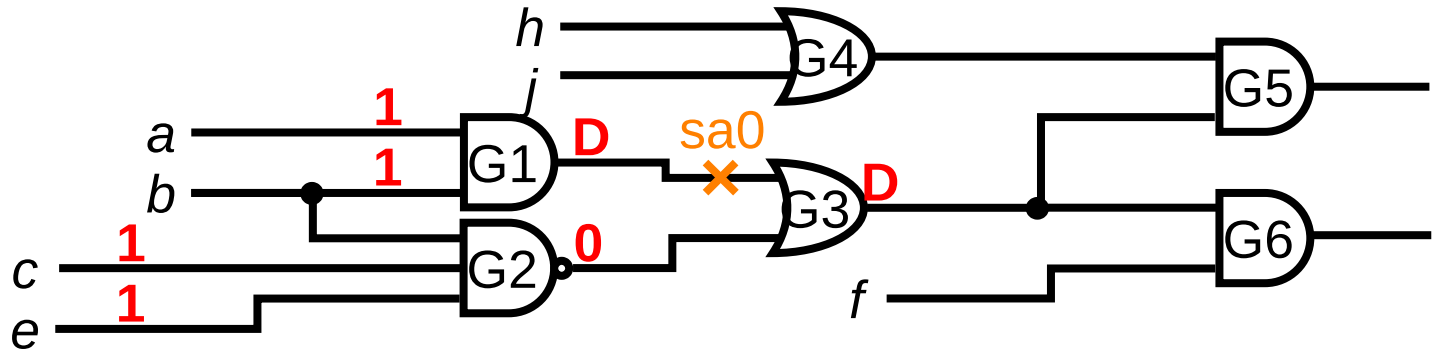
- 1. Create D-frontier (fault activation)
- 2. **Drive** D-frontier toward output (fault effect propagation)
- 3. **Justify** J-frontiers
- 4. **Backtrack** if any conflict occurs



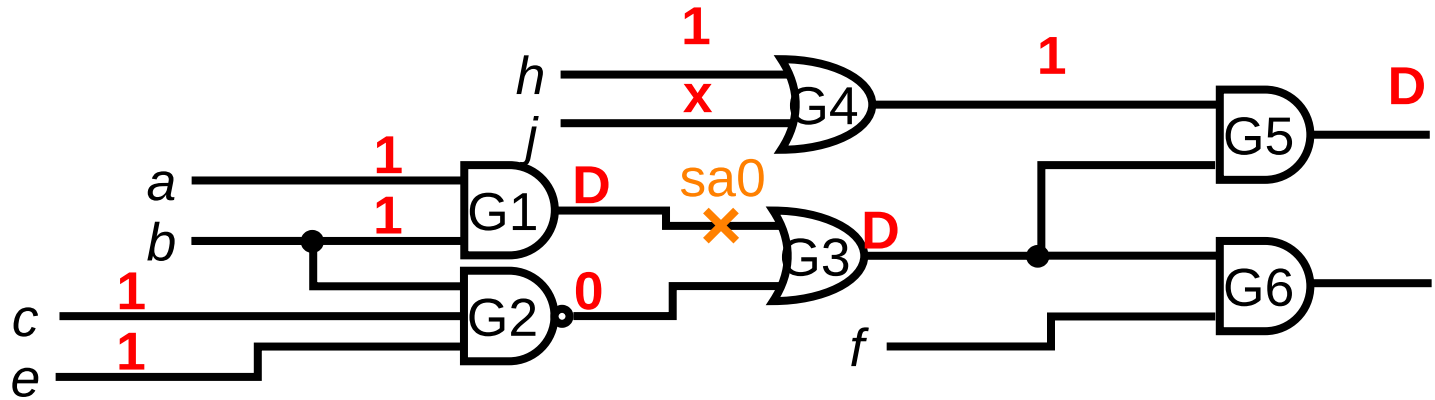
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Example



	a	b	c	e	G_1	G_2	G_3	h	j	G_4	f	G_5	G_6	Comments
Initial test cube $TC(0) =$ PDFC for G_1 stuck-at 0	1	1			D									Fault Activation. Implication: nothing happen.
Prop. D-cube of G_3 , PD_{G_3}					D	0	D							D-frontier: {G3}
$TC(1)=TC(0)\cap PD_{G_3}$	1	1			D	0	D							D-Drive through G3
Singular Cover SC_{G_2}		1	1	1		0								
$TC(2)=TC(1)\cap SC_{G_2}$	1	1	1	1	D	0	D							Backward implication



	a	b	c	e	G ₁	G ₂	G ₃	h	j	G ₄	f	G ₅	G ₆	Comments
TC(2)	1	1	1	1	D	0	D							D-frontier: {G ₅ , G ₆ }
Propagation D-cube of G ₅ , PD _{G₅}							D			1		D		Choose path through G ₅
TC(3)=TC(2)∩PD _{G₅}	1	1	1	1	D	0	D			<u>1</u>		D		D-drive through G ₅ . D reach PO.
Singular Cover SC _{G₄}								1	X	1				Justification J-frontier = {G ₄ }
TC(4)=TC(3)∩SC _{G₄}	1	1	1	1	D	0	D	1	X	1	X	D	X	Done.

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Primitive D-Cubes for a Fault (PDCF)

- Specify minimal input conditions
 - ♦ applied to gate input to produce error at gate output
 - ♦ Used in *fault activation* (more on this later)
- Example: AND gate output stuck-at faults:
 - ♦ Stuck-at-0 fault: 11D
 - ♦ Stuck-at-1 fault: 0xD' and x0D'



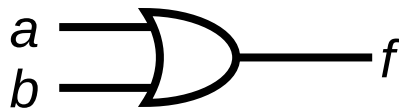
Singular Cover (SC)

- Minimum gate input assignments for gate output =0 or =1
 - Used in line *justification* or *implication* (more on this later)



Singular covers of AND

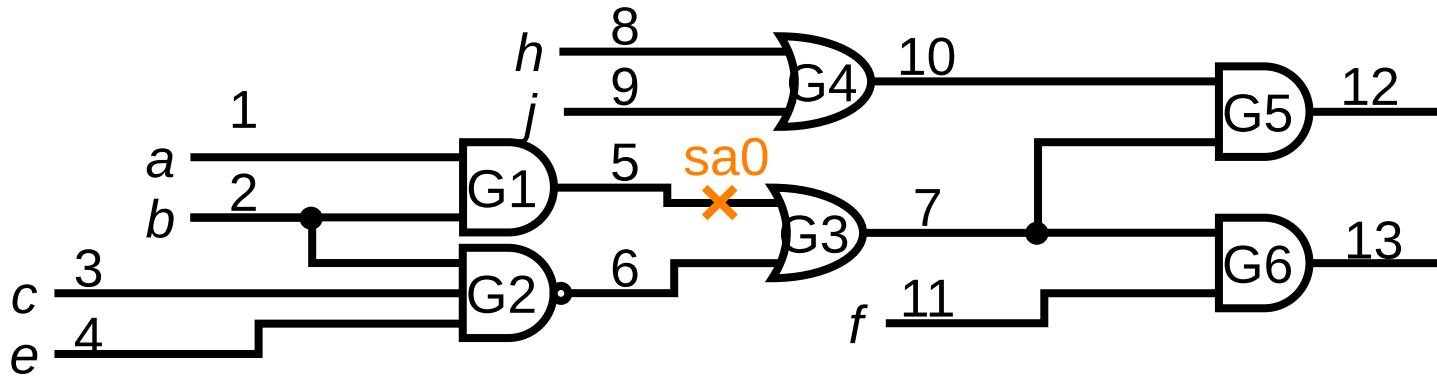
a	b	f
1	1	1
0	x	0
x	0	0



Singular covers of OR

a	b	f
0	0	0
1	x	1
x	1	1

Example



	a	b	c	e	G ₁	G ₂	G ₃	h	j	G ₄	f	G ₅	G ₆
Primitive D-cube for G1 sa0 fault	1	1			D								
Singular covers of G2		0 x x 1	x 0 x 1	x x 0 1		1 1 1 0							
Singular covers of G3					1 x 0	x 1 0	1 1 0						
Singular covers of G4								1 x 0	x 1 0	1 1 0			
Singular covers of G5							x 0 1			0 x 1		0 0 1	
Singular covers of G6							x 0 1				0 x 1		0 0 1

Propagation D-Cube (PDC)

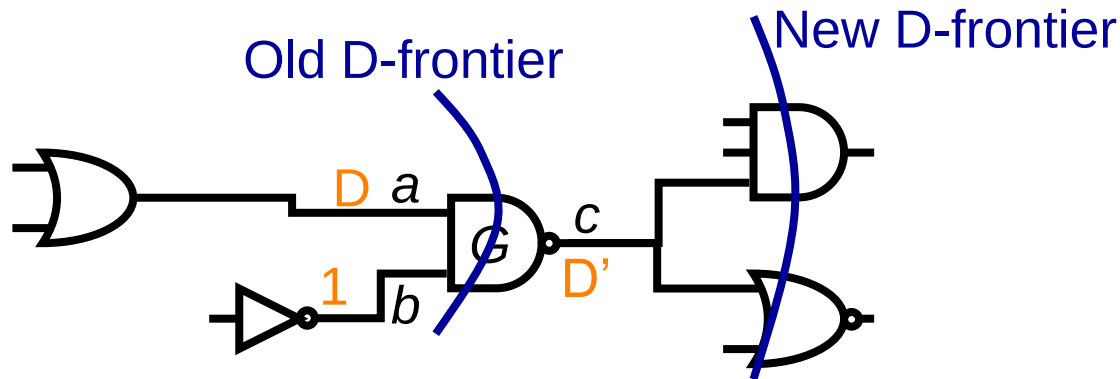
- Minimum gate input assignments required
 - ♦ to propagate a D or D' from gate input(s) to gate output
- Used in *D-Drive* or *implication* (more later)

<i>A</i>	<i>B</i>	<i>AB</i>
D	1	D
1	D	D
D	D	D
D'	1	D'
1	D'	D'
D'	D'	D'

<i>A</i>	<i>B</i>	$(A+B)'$
0	D	D'
D	0	D'
D	D	D'
0	D'	D
D'	0	D
D'	D'	D

D-drive

- D-drive selects an element in D-frontier
 - ◆ and attempts to propagate D or D' from gate input to gate output
 - ◆ Using propagation D-cube



	...	a	b	c	...
Test cube before D-drive	...	D	X	X	
Propagation D cube of gate G	...	D	1	D'	...
Test cube after D-drive	...	D	1	D'	...

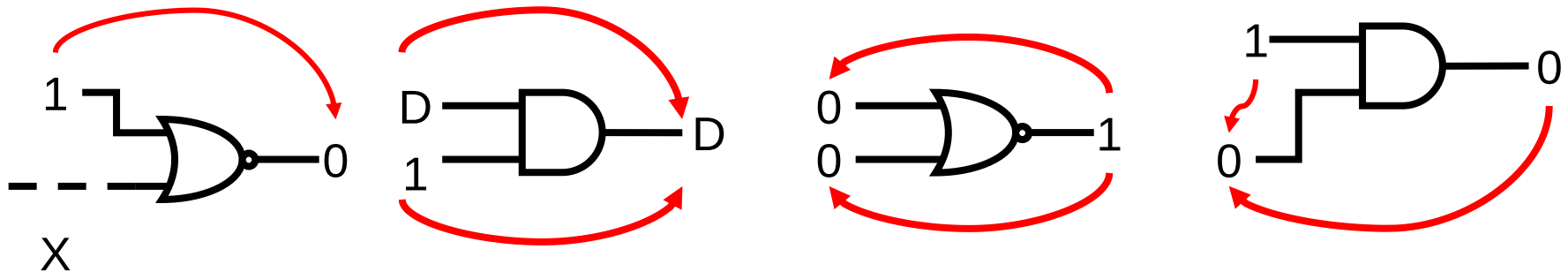
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Implication

- **Forward Implication**
 - ♦ partially (or fully) specified input values *uniquely* determines the output values.
- **Backward Implication**
 - ♦ knowing the output values (and some input values) can *uniquely* determine the un-specified input values.

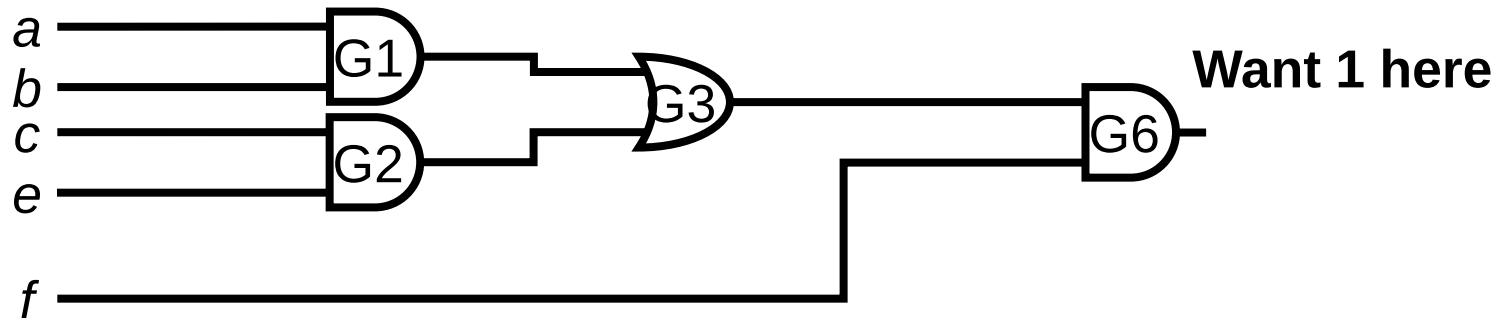
- Examples



- **Note 1: Implication means NO choice**
- **Note 2: Implication can be done any time a decision is made**

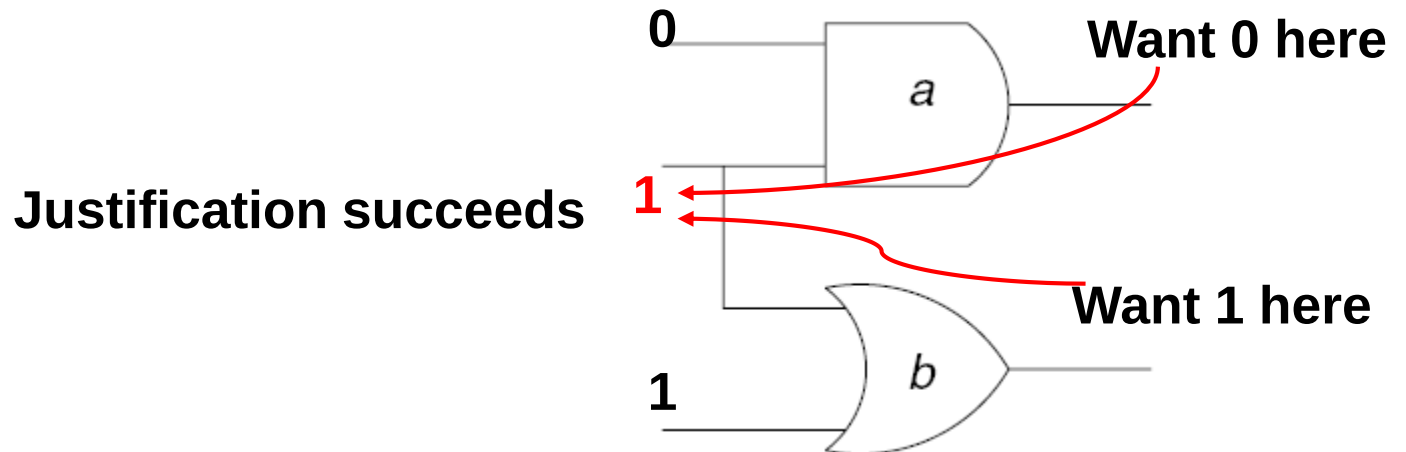
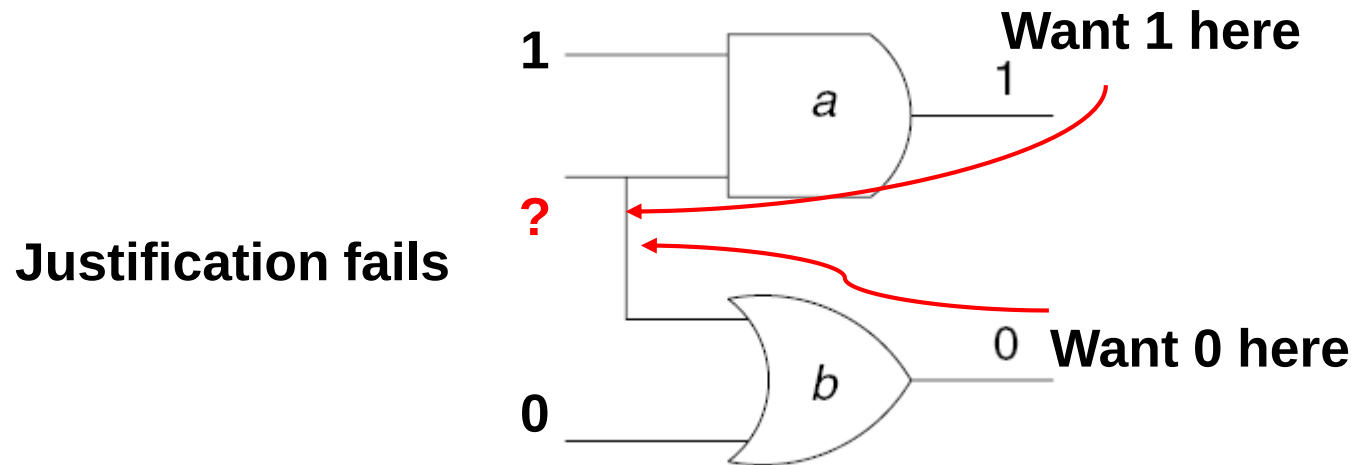
Justification

- Definition: find a valid primary input assignment for desired values
- Justification is easy inside a *fanout-free circuit*
- No decision needed
 - ♦ Always finds an answer
- Example



Justification (2)

- Justification may fail when there are **fanout branches**



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Test Cube (TC)

- A **Test cube** is a partially specified Boolean values for testing a fault
 - In D algorithm, a test cube contains
 - ♦ not only primary inputs, but also internal nodes
 - Notation
 - ♦ **TC(n)** = test cube at ATPG step n
-
- NOTE: In PODEM, a test cube contains primary inputs only
 - * More details later

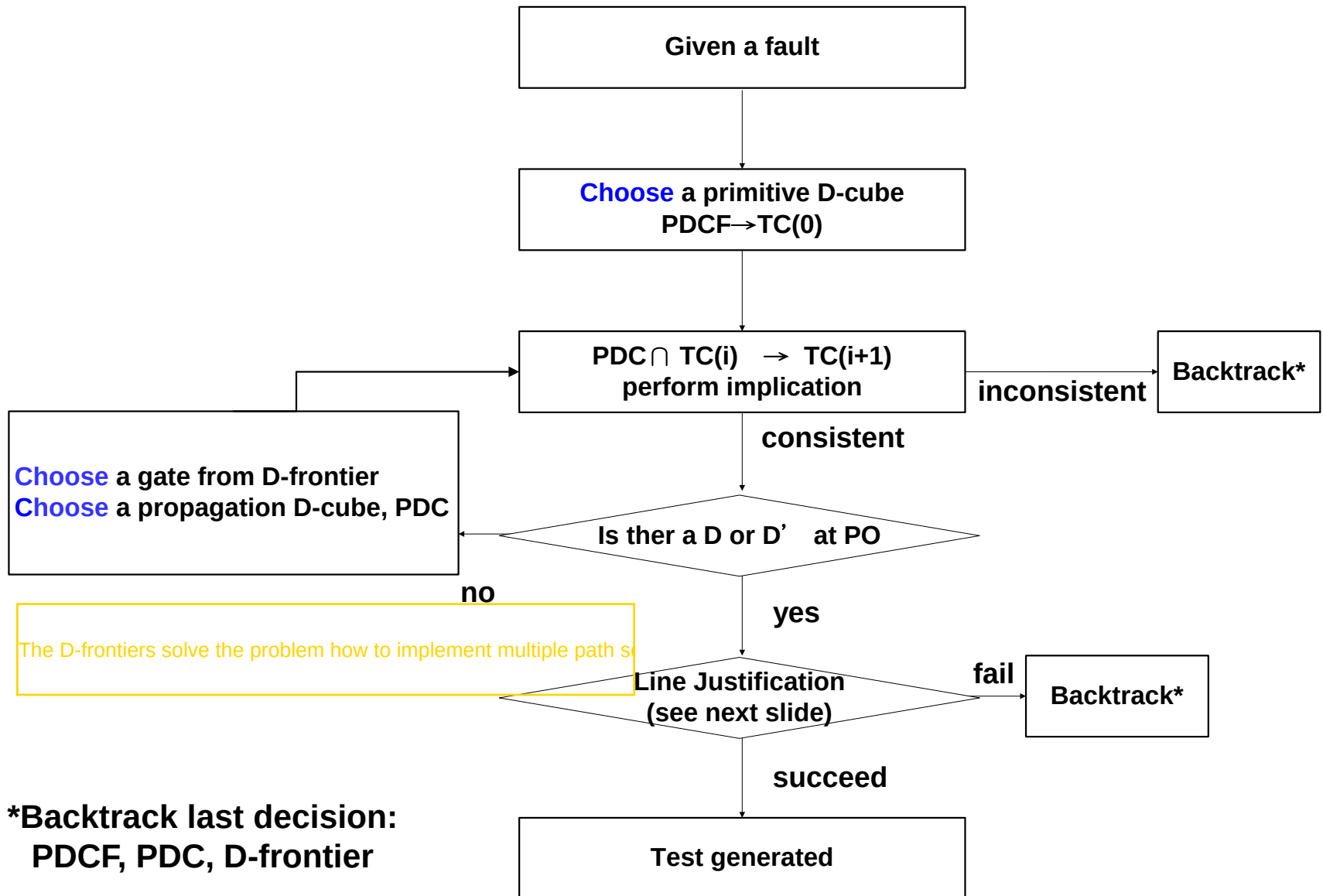
Intersection of Test Cubes

- Two bits has *intersection* if their logic values are not conflicting
- Two test cubes has intersection if there is no confliction in any bit
- Example :
 - ♦ $TC(1) \cap TC(2) = X0X1 \cap 1XXX = 10X1$
 - ♦ $TC(1) \cap TC(2) = 10X1 \cap 0XXX = \text{no intersection}$

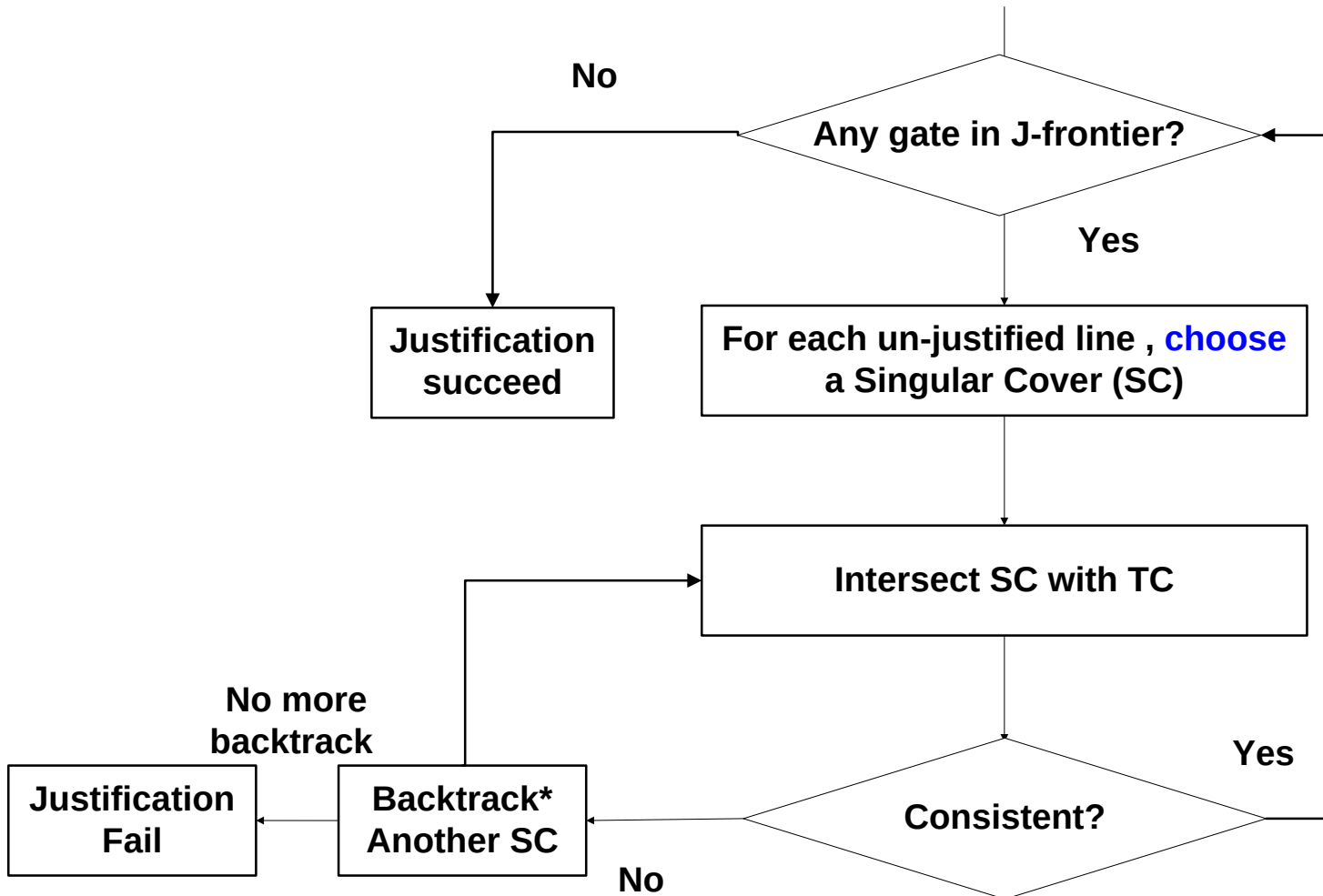
Bit 1 $\cap$ Bit 2					
bit2 \ bit1	0	1	X	D	D'
0	0		0		
1		1	1		
X	0	1	X	D	D'
D			D	D	
D'			D'		D'

empty box = confliction = no intersection

Flowchart of the D-Algorithm



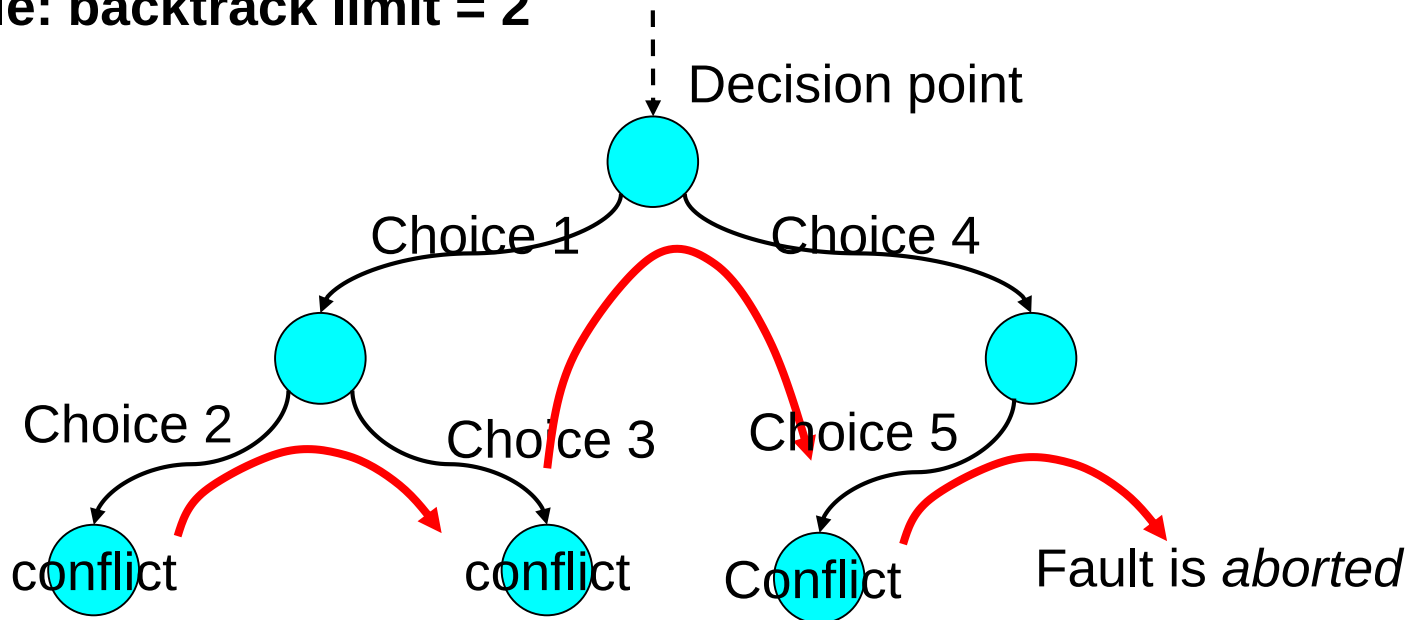
Line Justification



***backtrack last decision: SC**

Backtrack

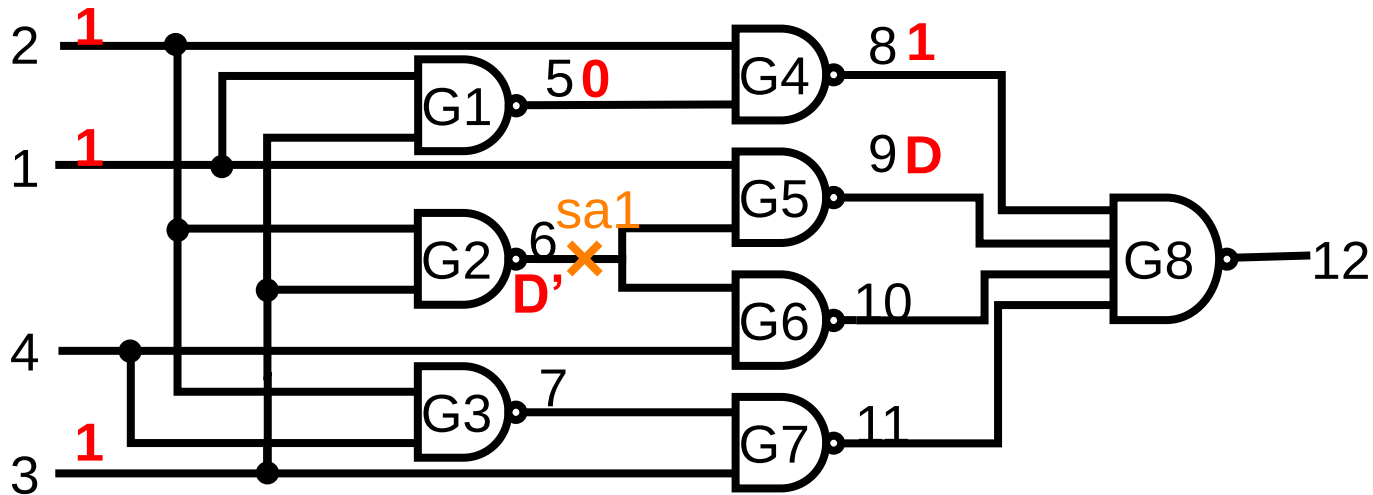
- When conflict, **backtrack** to last decision point and change choice
 - ♦ Choice can be: PDCF, PDC, SC, D-drive gate
- To avoid spending too much time on a fault
 - ♦ Use specify a **backtrack limit**
 - * maximum number of backtracks allowed for a single fault
 - ♦ Fault is **aborted** if backtrack limit is reached
- Example: backtrack limit = 2



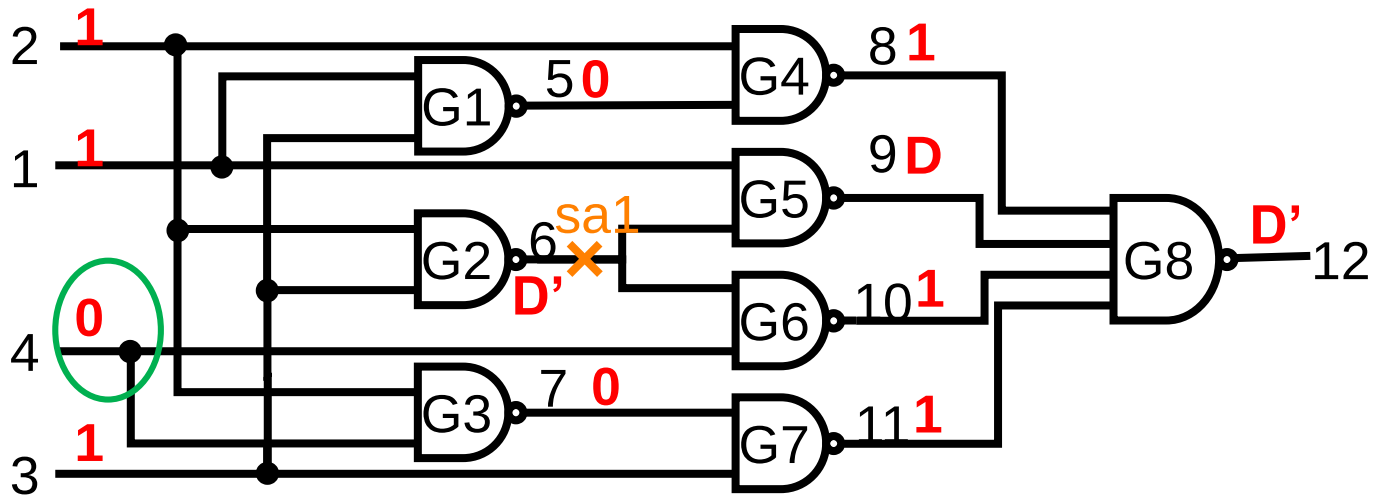
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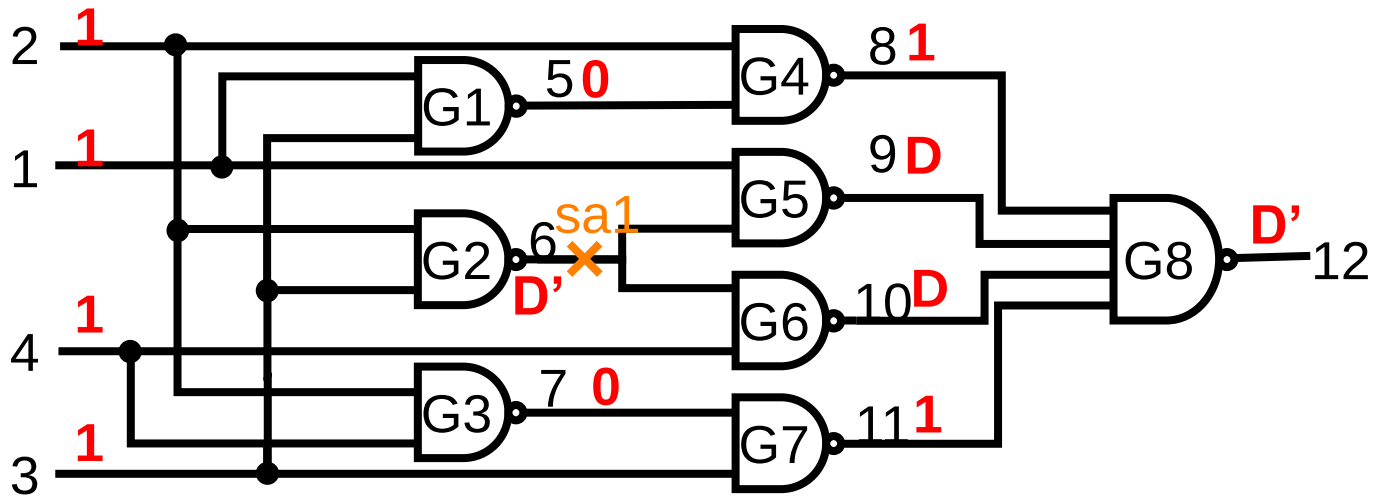
Another D-Algorithm Example



		1	2	3	4	5	6	7	8	9	10	11	12
1	PDCF:TC(0) = PDFC		1	1			D'						
2	Implication: nothing happens		1	1			D'						
3	choose D-frontier G5: PDC TC(1)=TC(0) ∩ PDC	1					D'			D			
		1	1	1			D'			D			
4	Forward Implication: SC _{G1} TC(2)	1		1		0							
		1	1	1		0	D'			D			
5	Forward Implication: SC _{G4} TC(3)		X			0			1				
		1	1	1		0	D'		1	D			



		1	2	3	4	5	6	7	8	9	10	11	12
	TC(3)	1	1	1		0	D'		1	D			
6	choose D-frontier G8: PDC TC(4)	1	1	1		0	D'		1	D	1	1	D'
7	Backward Implication G ₇ TC(5)	1	1	1		0	D'	0	1	D	1	1	D'
8	Backward Implication G ₆ : SC _{G6} TC(6)	1	1	1	0	0	D'	0	1	D	1	1	D'
9	Backward Implication G ₃ : SC _{G3} Fail. Backtrack step to 6		1		1			0					



		1	2	3	4	5	6	7	8	9	10	11	12
	Backtrack to step 6. TC(3)	1	1	1		0	D'		1	D			
10	Choose D-frontier G6: PDC TC(4)	1	1	1	1	0	D'		1	D	D		
11	Forward Implication: SC _{G3} TC(5)	1	1	1	1	0	D'	0	1	D	D		
12	Forward Implication: SC _{G7} TC(6)	1	1	1	1	0	D'	0	1	D	D	1	
13	Forward Implication: SC _{G8} TC(7)	1	1	1	1	0	D'	0	1	D	D	1	D'
14	No justification needed. Test generated.	1	1	1	1	0	D'	0	1	D	D	1	D'

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- Plus & Minus of D-Algorithm

Plus/Minus of D-Algorithm

- + D algorithm is *complete ATPG*
 - ◆ Guarantee to generate a pattern for a testable fault
- Large search space
 - ◆ Assignment of values is allowed for internal signals
 - ◆ Backtracking could occur at each gate
 - ◆ Very large search space

FFT

- **Q1. Why we do justification at the end of D-algorithm**
 - ♦ **Why not immediately after implication?**
- **Q2. Please explain why D-algorithm is complete ATPG**